

TASK 03 – Prompting for Task Automation

Prompt Engineering Internship

In this task, prompting is used to automate the process of extracting structured information from unstructured text. A reusable prompt is designed to convert different types of task-related notes into a consistent JSON format. The prompt is tested with three different inputs to check whether it can produce reliable and structured outputs.

Automation Task: Converting Notes into Structured JSON

Reusable Automation Prompt

Prompt:

You are a data extraction assistant. Read the given text and extract the available information into a valid JSON object. Use the following fields: **name, task, deadline, priority, status**. If any information is missing, use "Not specified" instead of guessing. Keep the original meaning of the input and return only valid JSON without additional explanation.

Input 1

Input:

Amrin needs to complete the Prompt Engineering Task 1 by August 25. It is a high-priority task and is currently in progress.

Output:

```
{  
  "name": "Amrin",  
  "task": "Prompt Engineering Task 1",  
  "deadline": "August 25",  
  "priority": "High",  
  "status": "In progress"  
}
```

Analysis:

The prompt correctly identifies all five required fields from the input and converts the natural-language sentence into a consistent JSON structure.

Input 2

Input:

Rahul has to submit the Python assignment on September 2. The assignment is currently pending and has medium priority.

Output:

```
{  
  "name": "Rahul",  
  "task": "Python assignment",  
  "deadline": "September 2",  
  "priority": "Medium",  
  "status": "Pending"  
}
```

Analysis:

The same prompt successfully handles a different person's task and extracts the required information without changing the meaning of the original text. This demonstrates that the prompt can be reused for multiple inputs.

Input 3

Input:

Sara is working on her database project. She needs to finish it next Monday.
The project is not started yet.

Output:

```
{  
  "name": "Sara",  
  "task": "Database project",  
  "deadline": "Next Monday",  
  "priority": "Not specified",  
  "status": "Not started"  
}
```

Analysis:

The prompt handles missing information correctly. Since no priority is mentioned in the input, it returns "Not specified" rather than creating an assumption. This makes the automated output more reliable.

Prompt Iteration and Debugging

During the initial prompt design, the model could potentially add explanations along with the extracted information. To improve reliability, the instruction **“return only valid JSON without additional explanation”** was added.

Another important improvement was adding the rule **“If any information is missing, use ‘Not specified’ instead of guessing.”** This prevents the model from generating unsupported information.

These small changes made the prompt more consistent and suitable for repeated use with different inputs.

Key Learnings from Task 03

- **Reusable Prompts** – A well-designed prompt can be applied to multiple inputs without rewriting it each time.
- **Consistent Output** – Specifying a fixed structure helps produce predictable results.
- **Handling Missing Data** – Clear instructions prevent the AI from guessing unavailable information.
- **Output Validation** – Asking for valid JSON makes the output easier to use in applications and automation workflows.
- **Prompt Debugging** – Testing the prompt with different inputs helps identify and correct unwanted behaviour.
- **Task Automation** – Prompting can reduce repetitive manual work by converting unstructured information into structured data.

Conclusion

This task demonstrates how prompt engineering can be used for practical task automation. A clear and reusable prompt can transform different natural-language inputs into a consistent structured format. By testing multiple examples and improving the prompt through iteration, the reliability and usefulness of the automated output can be increased.